



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



GNSS Interference Aviation Operational Cyber Risk Daily Review

Threat Report

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TLP:CLEAR | Lost Boys Cyber | GNSS Interference Aviation Operational Cyber Risk Daily Review - Threat Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

Table of Contents

1. Executive Summary
2. Intelligence Requirements
3. Source Review & Confidence
4. Threat Activity Overview
5. Affected Sectors and Exposure
6. Aviation and Aerospace Sector Impact Analysis
7. Tactics, Techniques, and Procedures
8. Infrastructure and Indicators
9. Detection Engineering
 - 9.1. Detection Summary
 - 9.2. Sigma / Detection Content
 - 9.3. Hunt Query
10. Threat Hunting
 - 10.1. Hunt Hypothesis
 - 10.2. Hunt Query
11. Risk Assessment
12. Recommended Actions
13. References

GNSS Interference Aviation Operational Cyber Risk Daily Review

1. Executive Summary

Aviation GNSS jamming and spoofing remain operational cyber-physical risks because navigation and timing dependencies can be degraded or manipulated. Aviation defenders should align cyber monitoring with flight operations reporting and vendor resilience plans.

This daily report is intended for defender prioritization rather than attribution finality. Source expansion beyond 24-72 hours was used where the best primary source remained recently relevant but was published earlier than today's run.

2. Intelligence Requirements

Question	Current Answer	Confidence
What changed that defenders should review today?	GNSS interference includes jamming, spoofing, and meaconing effects that can degrade avionics trust and connected operational systems. While not always an IT intrusion, it creates cyber-physical consequences and resilience requirements.	Medium
Which environments should care most?	airlines, business aviation, airports, avionics/MRO, flight operations.	Medium
What should the SOC do first?	Validate exposure, run the included telemetry hunts, and confirm response owners for likely impact paths.	High

3. Source Review & Confidence

Source	Contribution	Confidence
ION abstract	Aviation GNSS interference threat discussion.	High
NBAA	Aviation operator mitigation context.	Medium
FAA GPS/GNSS interference resource	Official aviation GNSS resource page.	Medium
AeroVigil	Sector explainer on jamming/spoofing.	Medium

4. Threat Activity Overview

GNSS interference includes jamming, spoofing, and meaconing effects that can degrade avionics trust and connected operational systems. While not always an IT intrusion, it creates cyber-physical consequences and resilience requirements.

The most important operational takeaway is not a single indicator; it is the behavior pattern and exposed dependency class. Defenders should use the detections and hunts below as starting points, then tune to local log schemas.

5. Affected Sectors and Exposure

Exposure Pattern	Why It Matters
Directly exposed technology or service	Increases likelihood of opportunistic targeting and automated probing.
Identity, email, remote-access, or collaboration dependency	Increases blast radius because successful access can bypass

	perimeter assumptions.
Supplier-managed or shared service	Creates concentration risk and incident-notification delays.
Operationally critical workflow	Converts cyber compromise into business or mission disruption.

6. Aviation and Aerospace Sector Impact Analysis

Sector Impact Field	Assessment
Affected segment	airlines, business aviation, airports, avionics/MRO, flight operations
Operational relevance	The issue can disrupt passenger processing, mission systems, satellite/space dependencies, regulatory oversight, or aviation operational continuity depending on local exposure.
Likely exposure paths	Shared IT providers, remote access, SaaS/identity platforms, supplier connectivity, operational technology, ground-segment systems, and connected avionics support environments.
Third-party / supply-chain considerations	Aviation and aerospace operators should require incident notification, continuity procedures, and cyber-control evidence from shared technology providers and critical suppliers.
Defender actions	Map dependencies, confirm backup/manual procedures, monitor identity and remote-access telemetry, and rehearse escalation between SOC, operations control, safety, and vendor-management teams.

7. Tactics, Techniques, and Procedures

Tactic / Phase	Observed Technique	Evidence
Initial Access	Phishing, exposed service, supply-chain, or remote-access abuse	Reflected in the cited reporting and sector exposure.
Execution / Persistence	Scripted tooling, malware, credential/session abuse, or provider-side compromise	Varies by campaign; validate with endpoint and application telemetry.
Defense Evasion	Use of legitimate tools, trusted platforms, or edge infrastructure	Common pattern across the reviewed reporting.
Impact	Data theft, service disruption, ransomware/extortion, or operational degradation	Primary defender concern for airlines, business aviation, airports, avionics/MRO, flight operations.

8. Infrastructure and Indicators

Indicator / Infrastructure	Type	Operational Use
No normalized hard IOC set in this automated review	Source limitation	Use behavior and exposure hunts; ingest vendor IOC bundles when available.
Rare external authentication sources	Behavioral indicator	Triage abnormal access into affected

		systems.
New or unusual outbound connections from edge/server workloads	Behavioral indicator	Hunt possible relay, malware, or exfiltration staging.

9. Detection Engineering

9.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
Rare admin or service login after advisory/report publication	Identity/app logs	Identify initial access or supplier compromise.
Unusual child processes from browsers, package managers, admin consoles, or service accounts	Endpoint telemetry	Catch execution following social engineering or supply-chain compromise.
New outbound connections to rare hosts from servers or build systems	Network/EDR	Surface C2, relay-node, or exfiltration behavior.

9.2. Sigma / Detection Content

```

title: Daily Threat Review Suspicious Post Access Behavior
status: experimental
description: Behavioral seed analytic for daily Lost Boys Cyber threat reports; tune service names and parent processes locally.
logsource:
  product: windows
  category: process_creation
detection:
  selection_parent:
    ParentImage|endswith:
      - '\chrome.exe'
      - '\msedge.exe'
      - '\node.exe'
      - '\npm.cmd'
      - '\powershell.exe'
  selection_child:
    Image|endswith:
      - '\powershell.exe'
      - '\cmd.exe'
      - '\wscript.exe'
      - '\rundll32.exe'
      - '\curl.exe'
  condition: selection_parent and selection_child
level: medium

```

9.3. Hunt Query

```

DeviceProcessEvents
| where Timestamp > ago(14d)
| where InitiatingProcessFileName in~ ("chrome.exe", "msedge.exe", "node.exe", "npm.cmd", "powershell.exe")
| where FileName in~ ("powershell.exe", "cmd.exe", "wscript.exe", "rundll32.exe", "curl.exe", "msiexec.exe")
| summarize events=count(), firstSeen=min(Timestamp), lastSeen=max(Timestamp) by DeviceName, InitiatingProcessFileName, FileName, ProcessCommandLine
| order by events desc

```

10. Threat Hunting

10.1. Hunt Hypothesis

If this activity is present in the environment, defenders will observe a combination of anomalous authentication, unusual process lineage, rare network destinations, and supplier or operational workflow disruptions.

10.2. Hunt Query

<pre>DeviceNetworkEvents where Timestamp > ago(14d) where InitiatingProcessFileName in~ ("powershell.exe", "cmd.exe", "node.exe", "npm.exe", "chrome.exe", "msedge.exe") summarize connections=count(), firstSeen=min(Timestamp), lastSeen=max(Timestamp), remotePorts=make_set(RemotePort, 10) by DeviceName, InitiatingProcessFileName, RemoteUrl, RemoteIP where connections > 3 order by connections desc</pre>
<pre>SigninLogs where TimeGenerated > ago(14d) summarize attempts=count(), failures=countif(ResultType != 0), countries=make_set(Location, 10) by UserPrincipalName, AppDisplayName, IPAddress where failures > 10 or array_length(countries) > 2</pre>

11. Risk Assessment

Risk Driver	Assessment
Likelihood	Medium for organizations matching the exposed technology/sector pattern.
Impact	Ranges from credential theft and data exposure to operational disruption or ransomware.
Confidence	Medium; local validation required before incident declaration.
Priority Environments	airlines, business aviation, airports, avionics/MRO, flight operations.

12. Recommended Actions

Priority	Owner	Action
Critical	SOC	Run the included hunts and tune for named platforms, suppliers, or services in scope.
High	IT / Platform owners	Confirm exposed assets, administrative access paths, and recent patch/configuration posture.
High	Vendor management	Identify third-party dependencies and request incident/control attestations where relevant.
Medium	IR lead	Prepare containment and communications playbooks for the most likely impact path.

13. References

- [1] ION abstract: <https://www.ion.org/publications/abstract.cfm?articleID=20513>

- [2] NBAA: <https://nbaa.org/news/business-aviation-insider/2026-07/3-emerging-technologies-to-outsmart-aircraft-gps-spoofing-jamming/>
- [3] FAA GPS/GNSS interference resource: https://www.faa.gov/air_traffic/flight_info/aeronav/gnss
- [4] AeroVigil: <https://aerovigil.com/blog/gnss-interference-gps-jamming-spoofing-aviation>